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The House Sparrow *Passer domesticus* in urban areas: reviewing a possible link between post-decline distribution and human socioeconomic status

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Abstract The House Sparrow *Passer domesticus* is traditionally associated with human habitation. However, the species has undergone dramatic declines in many urban areas in north-western Europe. There are many theories as to why this decline has occurred, but the lack of data on House Sparrow numbers prior to their decline has hampered efforts to investigate these theories in detail. This review summarises the demographic changes in urban House Sparrow populations since the 1970s, and considers evidence that the current distribution of House Sparrows may reflect changes in urban habitats caused by socioeconomic change. Evidence is mounting that, within urban landscapes, House Sparrows appear to be more prevalent in areas with a relatively low human socioeconomic status. Here, we present evidence to suggest that House Sparrows may have disappeared predominantly from more affluent areas, and that these areas are more likely to have undergone changes to habitat structure. We also show how these changes in habitat could influence House Sparrow populations via impacts upon nesting success, foraging and predation risk.

Keywords Brownfield sites · Deprivation · Habitat · Regeneration

Introduction

As a landscape becomes more urbanised, avian population density often increases. However, species diversity tends to decrease in comparison to other habitat types (Beissinger and Osbourne 1982; Blair 1996; Marzluff et al. 2001). The House Sparrow (*Passer domesticus*, Linnaeus) is traditionally associated with human habitation, and is one of the few species that generally persists as urbanisation increases. It is a widespread species, but in recent years there has been an abrupt decline in House Sparrow numbers in urban areas throughout north-western Europe, the cause of which is not known at present, although several hypotheses have been suggested (Summers-Smith 2003).

House Sparrows were so common they were regarded as a pest species during the eighteenth and nineteenth centuries (Crick 2002), particularly in towns and cities. However, populations have been declining in many urban areas since the mid-1980s (Crick 2002). In the UK, this trend is particularly apparent in the south and east, which hold just over 50% of the sparrow population (Siriwardena et al. 2002). The overall population of UK House Sparrows is estimated to have declined from approximately 13 million pairs in the early 1970s to around 6 million pairs by the late 1990s (Crick and Siriwardena 2002; Robinson et al. 2005). This decline is so widespread that the House Sparrow is currently on the red list reserved for species of special conservation concern in the UK (Gregory et al. 2003). Similar declines have been reported in urban areas throughout north-western Europe (e.g. Prowse 2002; Mulsow 2005, 2006; Stastny et al. 2005; Heij 2006). As a result, the House Sparrow is

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now listed as a species of European conservation concern (SPEC category 3; Birdlife International 2006). Although little historical data is available, population monitoring of House Sparrows has increased since the 1970s, and a more detailed picture of the state of the House Sparrow in Europe is beginning to emerge.

Here, we review the demographic changes to House Sparrow populations in the UK, and summarise the changes to populations in European cities where data are available. We discuss recent evidence suggesting that the current distribution of House Sparrows in many urban areas is linked to human socioeconomic status, and consider potential mechanisms to explain a possible link between House Sparrow populations and social deprivation. We focus on three ways in which socioeconomic status could influence House Sparrow populations: by creating differences in habitat structure which may impact upon foraging success; by causing indirect effects such as increased predation risk; and potential effects on nest site availability. Finally, we consider how urban regeneration and redevelopment may affect House Sparrow populations on a city-wide scale.

Demographic changes

Population monitoring in the UK has provided a relatively detailed picture of House Sparrow numbers. A number of surveys have shown declines both in the wider countryside (Marchant et al. 1990; Robinson et al. 2005; Raven and Noble 2006) and especially in private gardens (Cannon et al. 2005; Chamberlain et al. 2005; Toms 2006) since the mid- to late 1970s. There is evidence that declines vary both geographically and according to habitat. Raven and Noble (2006) showed that declines are occurring in most urbanised parts of England, whereas in Scotland and Wales, populations are increasing. Furthermore, Chamberlain et al. (2005) suggest that declines started earlier in suburban gardens than rural gardens. The timing of the decline in urban areas and the sedentary nature of House Sparrows would suggest that urban and rural populations are relatively isolated from each other, and that the urban population decline is not a result of decreased immigration from farmland areas as a result of agricultural intensification (Siriwardena et al. 2002). House Sparrow populations within a city are likely to overlap more than those in rural areas, as they exist in closer proximity to each other. However, adult House Sparrows rarely disperse over distances of more than 2 km even outside the breeding season (Snow and Perrins 1998). A genetic study of four rural House Sparrow populations showed very restricted movement between populations (Hole et al. 2002), which suggests that rural House Sparrows are unlikely to be a source of recruitment to the urban breeding population.

Data from long-term UK-wide nest monitoring under the Nest Record Scheme (NRS) show that nest failure rates in House Sparrows were higher in urban/suburban than rural habitats (Crick 2002). However, a caveat on the interpretation of NRS data is that measures are only per nesting attempt, rather than over a whole season. If, for example, a lack of food led to fewer nesting attempts and lower overall productivity per breeding season, this would not be detected. This, and differences in population growth rates in relation to breeding performance, suggest that the mechanisms driving population trends in rural and urban/suburban habitats differ, and it is therefore appropriate to consider the two habitats separately.

Detailed data on House Sparrow populations in other European countries are not as readily available as for the UK. However, population monitoring in individual cities has also provided evidence for a broad scale population decline. This appears to be the case for many cities across north-western Europe, although in others no evidence of a decline has been noted. Table 1 provides a summary of cities for which trends in the House Sparrow population are available.

A link to socioeconomic status

The distribution of House Sparrows within urban areas is often very patchy, and recent evidence suggests that human socioeconomic status may provide a partial explanation for

Table 1 A summary of reported House Sparrow *Passer domesticus* population trends in European cities

City	Overall trend	Reference
Berlin	Stable	Witt (2000, 2005)
Bristol	Decline (pre-1994)	Bland (1998)
Brussels	Stable	Weiserbs and Jacob (2005)
Dublin	Decline	Prowse (2002)
Edinburgh	Decline	Dott and Brown (2000)
Hamburg	Decline	Mulsow (2005, 2006)
Lisbon	Increase	Geraldes and Costa (2005)
London	Decline	Sanderson (1996)
Manchester	Stable	Prowse (2002)
Moscow	Decline	Kostantinov and Kakhnov (2005)
Norwich	Decline	Paston (2000)
Paris	Stable	McCarthy (2006)
Prague	Decline	Stastny et al. (2005)
Rotterdam	Decline	Heij (2006)
St Petersburg	Decline	Khrabryi (2005)
Warsaw	Slight decline	Luniak (2005)

this. It appears that the remaining House Sparrow colonies are biased towards relatively deprived areas in many cities. For example, in Bristol, UK, House Sparrows are still common in the most deprived areas of the city as measured by the English Indices of Deprivation (Noble et al. 2004), but they are almost completely absent from areas which have a higher socioeconomic status (Bland 1998). Similarly, in Norwich, most remaining sparrow colonies are to be found on council estates in the north of the city (Paston 2000). Evidence from Paris and Berlin also links House Sparrows with deprived areas (Witt 2000, 2005; Malher, unpublished data). If this link is confirmed, it is likely that changes to areas with a high socioeconomic status have contributed to the decline of sparrow populations in those areas, whilst lower status areas have been relatively unaffected. Recent evidence suggests that areas of varying socioeconomic status do experience different levels of habitat change over time (Pauleit et al. 2005). Possible changes in habitat, and their effects on biodiversity are considered below, with particular reference to the potential impacts on House Sparrow populations in areas of differing socioeconomic status.

Garden management

Private gardens provide substantial resources for urban birds. However, in recent years an increase in home improvements and demand for off-road parking has led to changes in habitat structure within gardens (Pauleit et al. 2005). The presence of rough grass, nettles and weedy patches is linked to increased bird diversity in urban areas (Chamberlain et al. 2007a), which suggests that tidy gardens with high proportion of paving are less suitable habitat for foraging birds. This may cause problems for adult House Sparrows in the breeding season, when insect-based food is required for nestlings. Whilst there have been few studies of urban arthropods (McIntyre 2000), Vincent (2005) found that House Sparrow fledging success was higher in areas where insect abundance was high, and chicks were less likely to starve when fed a high proportion of insectivorous as opposed to vegetable food. This study also showed that urban House Sparrows have more success in fledging young in areas containing a high proportion of deciduous shrubs and relatively little concrete.

It is likely that gardens in more deprived areas will consist of more suitable habitat for House Sparrows than those in areas with a higher socioeconomic status. Relatively deprived areas are likely to contain a high proportion of native shrubs, and traditional council estates provide large blocks of garden habitat (especially back-to-back terraced housing) which can provide cover and insects for foraging birds. Evidence from areas such as Norwich, UK, appears to support this theory (Paston 2000). However, in

more affluent areas, the planting of predominantly ornamental shrubs and increased demand for off-street parking is likely to reduce the amount of habitat available to House Sparrows. In London, for example, it is estimated that around two-thirds of front gardens are now at least partially covered by paving as opposed to vegetation (London Assembly 2005a). A recent study of land cover change in Merseyside, UK, noted that loss of front gardens in order to provide paved parking areas was of particular concern in more affluent areas (Pauleit et al. 2005).

Indirect effects of habitat structure

Reducing the amount of foraging habitat available to House Sparrows is also likely to have indirect effects on survival and productivity. Factors such as predation and pollution are likely to affect urban birds at a city-wide scale, but may have a greater impact in areas of high socioeconomic status, where loss of greenery is already a problem. The interaction between predation risk and habitat structure has been documented by Whittingham and Evans (2004) in farmland habitats. This paper suggests that a mosaic of short and long grass may provide the optimal habitat to maximise foraging efficiency. Urban gardens also have a mosaic-like structure. However, the trend towards replacing native shrubs with paving or concrete (London Assembly 2005a) could cause the habitat structure of some areas to become more homogeneous, thereby increasing predation risk for foraging birds.

Sparrowhawks *Accipiter nisus* are common predators in urban habitats (Chamberlain et al. 2007b). House Sparrows appear to be especially vulnerable to predation by Sparrowhawks (Götmark and Post 1996), but there is no evidence that declines in House Sparrow abundance in urban gardens are linked to the recent population increase of Sparrowhawks in the UK (Chamberlain et al., in preparation). Domestic cats *Felis catus* are also capable of having a substantial negative impact on House Sparrow populations, and this has been identified as a possible factor in their decline (Churcher and Lawton 1987; Woods et al. 2003). The UK population of cats was estimated to be capable of predating 27 million birds in a 5-month period during 1997 (Woods et al. 2003), and a recent study in Bristol by Baker et al. (2005) found that House Sparrows were among the few species where predation rates by cats were high enough to have a detrimental effect on the local population. In areas where individuals are under increased foraging pressure due to suboptimal habitat, the risk of predation by cats or other birds may increase enough to affect population levels. Even if survival of adult birds is not immediately affected by increased predation risk, the survival of juvenile birds may be affected if adults spend more time avoiding predation.

Evidence from previous studies suggests that nest failure rates are high in some urban areas due to increased foraging or predation pressure. Although House Sparrow nest failure rates declined between 1975 and 2000 (Crick 2002), overall failure rates were significantly higher in urban/suburban habitats. This could indicate greater predation pressure in these habitats, but information on the cause of failures (e.g. predation, starvation or other factors) is not available. In addition, evidence from a study of House Sparrows in Leicester, UK, showed that the weight of chicks just prior to fledging was relatively low (Vincent 2005), a factor which is known to be linked to decreased post-fledging survival in a number of passerines (e.g. Garnet 1981; Magrath 1991). This, combined with a lack of food in late summer, could mean that recruitment of young birds to the breeding population is very low in some urban areas, particularly those with a low proportion of greenery.

It should also be noted that there may be more subtle effects of increased predation pressure that may alter birds' behaviour, including habitat selection (Suhonen 1993), flocking tendency in winter (Székely et al. 1989), and fat loading (Gosler et al. 1995). A recent study by MacLeod et al. (2006) found that adult House Sparrows in the UK appear to limit their uptake of fat in winter, suggesting they may be subject to mass-dependent predation effects. Increased predation may therefore have subtle and indirect effects that are not detectable through simple correlative studies, but may be enough to affect the survival and future reproductive success of individuals, particularly in suboptimal habitat. In areas with a low level of social deprivation, it is therefore possible that loss of vegetation in urban gardens may increase predation risk to a level that the population cannot sustain.

Nest site availability

House Sparrows often nest in the roof cavity of buildings, either under the tiles themselves or in and around the soffits and fascias. There is evidence to suggest that certain tile types and building methods may provide more suitable nesting sites for sparrows than others, and that these more suitable nesting spaces may be more abundant in areas with a low socioeconomic status. A recent small-scale survey of House Sparrow nests in Bristol, UK, found that they were strongly associated with roof tile types that are curved, as these provide large holes which are more accessible to sparrows (J. Tully, personal communication). In addition, plastic is replacing wood and cast iron as the material commonly used for soffits and fascias of houses. As wooden fascia boards rot, they create holes that can be used by nesting birds. Replacing rotten wooden boards with hardwearing plastic ones may lead to a substantial decrease in the number of nest holes available on a city-wide scale.

Therefore older houses, or those in a worse state of repair, are likely to be more suitable as nesting sites.

Evidence linking breeding House Sparrows to houses in poor condition exists in a number of urban areas. A nationwide survey of UK houses found that older (pre-1919) houses were important for hole-nesting birds such as sparrows, and that houses which had recently had repairs to the roofs were less likely to be associated with nesting birds (Wotton et al. 2002). House Sparrows appeared to avoid newer (post-1985) buildings, and were also much less likely to occur in houses that were built post-1945, and those that had undergone roof repairs in the last 10 years. In Harwich, UK, House Sparrows are also associated with post-war housing, but not with newer developments (Mason 2006). Less information is available for European cities, but in Vienna and Valencia, House Sparrows are also associated with damaged and neglected buildings (Murgui 2005; Sziemer and Holzer 2005). House Sparrows in Vienna are seen in old and new developments, but breeding sparrows are commonly associated with the older buildings (Sziemer and Holzer 2005). As neglected buildings are associated predominantly with areas of low socioeconomic status, it appears that these areas may provide more opportunities for nesting sparrows than areas which have newer buildings or those which have undergone repairs or regeneration. The loss of sparrow colonies from areas following regeneration has been recorded in Berlin (Witt 2000, 2005), which appears to support a possible link between changing socioeconomic status and House Sparrow population decline.

The wider urban environment

The loss of brownfield sites and green space from urban areas may exacerbate pressure on House Sparrows caused by loss of garden habitat in areas of high socioeconomic status. In addition, increasing levels of development and regeneration in many urban areas are likely to improve the socioeconomic status of human residents, but decrease the available habitat for House Sparrows. The London Assembly (2005b) defines brownfield sites as "previously developed land....currently or previously occupied by a permanent dwelling". These sites are often very rich in invertebrates and plants, and provide islands of cover in an urban landscape (Small et al. 2002; Eyre et al. 2003; Mason 2006; Muratet et al. 2007). The UK government is currently very keen to promote the use of these brownfield sites for development, and in 2001, 61% of new developments were built on brownfield sites (Office of the Deputy Prime Minister 2003). Similarly, in Paris, the proportion of such land has decreased by 24% in the last 20 years (Muratet et al. 2007). Regeneration, whilst likely to improve socioeconomic conditions for human inhabitants,

is therefore also reducing the amount of foraging habitat available to urban House Sparrows. This is likely to lead to colony loss in areas where human socioeconomic conditions improve.

Green space that has not been previously developed is also under threat in many urban areas due to regeneration and “infilling”, which increases housing density due to the purchase of private gardens for development. Loss of green space in Merseyside by this means was greater in more affluent areas between 1975 and 2000 (Pauleit et al. 2005), when the urban House Sparrow population also declined. However, the current demand for housing suggests that areas where socioeconomic conditions are relatively poor—and therefore where sparrow numbers are potentially high—may become targets for regeneration. A recent study of randomly selected urban sites throughout the UK identified residential areas and allotments as key habitats for House Sparrows, but predicted that a small increase in the proportion of flats (as opposed to houses with gardens) would be enough to cause rapid declines in House Sparrow abundance (Chamberlain et al. 2007b). Allotments in particular are prime targets for development in London (London Assembly 2006). New housing developments are currently being built at an average density of 22 dwellings per hectare (Entec UK 2004), and this is planned to increase to around 30 dwellings per hectare in areas such as the south east, where pressure on land is greatest (Office of the Deputy Prime Minister 2003). This is likely to reduce the foraging habitat available to sparrows in many areas undergoing redevelopment. If the trend towards increased housing density in urban areas continues, it is therefore likely that rapid declines in the House Sparrow population will continue in areas where redevelopment is taking place.

Redevelopment may also reduce the number of potential nest sites available even if the number of houses increases, as modern roofs are likely to provide fewer cavities than older ones or those in poor condition. It is also worthwhile to note, however, that House Sparrow populations may be lost even in areas where there are nesting opportunities and resources available. As sparrows are colonial nesters, a small decrease in the size of a colony may inhibit breeding in the remaining individuals. This could lead to the loss of the colony altogether (Summers-Smith 2003). This phenomenon is known as the Allee effect, and has been documented in a number of species (Veit and Lewis 1996). In the case of House Sparrows, the increasingly patchy distribution of the remaining urban colonies in many cities has increased the probability that this effect may occur and exacerbate any decline. Once a breeding colony has been lost, the sedentary nature of House Sparrows may also make it difficult for recolonisation to occur. Redevelopment of pockets of land in areas of low socioeconomic status could therefore affect adjacent House Sparrow

colonies, by reducing the overall number of nesting sites or foraging opportunities below the level at which the colonies can be sustained.

To date, attempts to determine the causes of decline in urban House Sparrow populations have been through relatively small-scale studies, as the complexity of urban habitats makes broad-scale studies difficult. We suggest that human socioeconomic status may provide a starting point for broader-scale analyses of the factors affecting urban sparrow populations. In particular, the importance of urban green space and brownfield sites should be considered in relation to foraging success and predation risk, and the structure and condition of houses considered in terms of nesting opportunities. This may allow for the development of mitigation measures in areas undergoing regeneration in the future. In addition, this approach could provide invaluable information to assist the planning of new settlements in order to create an urban landscape that is more sympathetic to the needs of the House Sparrow.

Zusammenfassung

Der Haussperling *Passer domesticus* in urbanen Regionen: Prüfen einer möglichen Verbindung zwischen seiner Verbreitung nach dem Rückgang der Bestände und humanem sozioökonomischem Status

Der Haussperling *Passer domesticus* wird traditionell mit menschlichen Wohnstätten in Verbindung gebracht. Die Art hat allerdings in vielen urbanen Gebieten Nordwesteuropas dramatische Bestandsrückgänge erfahren. Es gibt viele Theorien, weshalb dieser Rückgang eingetreten ist, doch das Fehlen von Daten zur Anzahl von Haussperlingen vor dem Rückgang hat eine detaillierte Untersuchung dieser Theorien erschwert. Diese Übersichtsstudie fasst die demographischen Veränderungen in urbanen Populationen des Haussperlings seit den 1970er Jahren zusammen und prüft Anzeichen dafür, dass die aktuelle Verbreitung von Haussperlingen Veränderungen in urbanen Habitaten, die durch sozioökonomischen Wandel verursacht werden, widerspiegeln könnte. Es gibt zunehmend Anhaltspunkte dafür, dass innerhalb urbaner Landschaften Haussperlinge in solchen Gegenden häufiger zu sein scheinen, in denen der humane sozioökonomische Status relativ niedrig ist. Hier führen wir Belege an, die darauf hindeuten, dass Haussperlinge vorwiegend aus reicheren Gegenden verschwunden sind und dass diese Gegenden mit höherer Wahrscheinlichkeit Veränderungen in der Habitatstruktur erfahren haben. Wir zeigen auch, wie diese Habitatveränderungen Populationen des Haussperlings durch Auswirkungen auf Nisterfolg, Nahrungssuche und Prädationsrisiko beeinflussen könnten.

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References

- Baker PJ, Bentley AJ, Ansell RJ, Harris S (2005) Impact of predation by domestic cats *Felis catus* in an urban area. *Mammal Rev* 35:302–312
- Beissinger SR, Osbourne DR (1982) Effects of urbanisation on avian community organisation. *Condor* 84:75–83
- Birdlife International (2006) Birds in Europe. Available from: <http://www.birdlife.org/datazone/species/BirdsInEuropeII/BiE2004/Sp8367.pdf>
- Blair RB (1996) Land use and avian species diversity along an urban gradient. *Ecol Appl* 6:506–519
- Bland R (1998) House sparrow densities in Bristol. *Avon Bird Rep* 1998:145–148
- Cannon AR, Chamberlain DE, Toms MP, Hatchwell BJ, Gaston KJ (2005) Trends in the use of private gardens by wild birds in Great Britain 1995–2002. *J Appl Ecol* 42:659–671. doi:10.1111/j.1365-2664.2005.01050.x
- Chamberlain DE, Vickery JA, Glue DE, Robinson RA, Conway GJ, Woodburn RJW, Cannon AR (2005) Annual and seasonal changes in the use of garden feeders by birds in winter. *Ibis* 147:563–575. doi:10.1111/j.1474-919x.2005.00430.x
- Chamberlain DE, Gough S, Vaughan H, Vickery JA, Appleton GF (2007a) Determinants of bird species richness in public green spaces. *Bird Study* 54:87–97
- Chamberlain DE, Toms MP, Cleary-McHarg R, Banks AN (2007b) House sparrow (*Passer domesticus*) habitat use in urbanized landscapes. *J Ornithol*. doi:10.1007/s10336-007-0165-x
- Churcher PB, Lawton JH (1987) Predation by domestic cats in an English village. *J Zool* 212:439–455
- Crick HQP (2002) Introduction. In: Crick HQP, Robinson RA, Appleton GF, Clark NA, Rickard AD (eds) Investigation into the causes of decline of starlings and house sparrows in Great Britain. BTO research report 290, DEFRA, Bristol. Available from: <http://www.defra.gov.uk/wildlife-countryside/resprog/findings/sparrow/>
- Crick HQP, Siriwardena GM (2002) National trends in the breeding performance of House Sparrows *Passer domesticus*. In: Crick HQP, Robinson RA, Appleton GF, Clark NA, Rickard AD (eds) Investigation into the causes of decline of starlings and house sparrows in Great Britain. BTO research report 290, DEFRA, Bristol. Available from: <http://www.defra.gov.uk/wildlife-countryside/resprog/findings/sparrow/>
- Dott HE, Brown AW (2000) A major decline in house sparrows in central Edinburgh. *Scott Birds* 21:61–68
- Entec UK (2004) Study into the environmental impacts of increasing the supply of housing in the UK. Available from: <http://statistics.defra.gov.uk/esg/reports/housing/mainrep.pdf>
- Eyre MD, Luff ML, Woodward JC (2003) Beetles (coleoptera) on brownfield sites in England: an important conservation resource? *J Insect Conserv* 7:223. doi:10.1023/B:JICO.0000021020.66549.1e
- Garnet MC (1981) Body size, its heritability and influence on juvenile survival among great tits, *Parus major*. *Ibis* 123:31–41
- Geraldes PL, Costa H (2005) Lisbon. In: Kelcey JG, Rheinwald G (eds) Birds in European cities. Ginster, Germany, pp 153–170
- Gosler AG, Greenwood JD, Perrins C (1995) Predation risk and the cost of being fat. *Nature* 377:621. doi:10.1038/377621a0
- Götmarm F, Post P (1996) Prey selection by sparrowhawks, *Accipiter nisus*: relative predation risk for breeding passerine birds in relation to their size, ecology and behaviour. *Philos Trans Biol Sci* 351:1559–1577
- Gregory RD, Eaton MA, Noble DG, Robinson JA, Parsons M, Baker H, Austin G, Hilton GM (2003) The State of the UK's birds 2002. Available from: <http://www.rspb.org.uk/ourwork/science/sotukb/index.asp>
- Heij K (2006) De Huismus *Passer domesticus*: achteruitgang, vermoedelijke oorzaken en oproep. *Vogeljaar* 54:195–207
- Hole DG, Whittingham MJ, Bradbury RB, Anderson GQA, Lee PLM, Wilson JD, Krebs JR (2002) Agriculture: widespread local house sparrow extinctions. *Nature* 418:931
- Khrabryi V (2005) St Petersburg. In: Kelcey JG, Rheinwald G (eds) Birds in European cities. Ginster, Germany, pp 307–334
- Kostantinov VM, Kakhanov R (2005) Moscow. In: Kelcey JG, Rheinwald G (eds) Birds in European cities. Ginster, Germany, pp 197–214
- London Assembly (2005a) Crazy paving: the environmental importance of London's front gardens. London Assembly Environmental Report, September 2005. Available from: <http://www.london.gov.uk/assembly/reports/environment.jsp>
- London Assembly (2005b) Dereliction of duty? A report on brownfield development in London. London Assembly Environmental Report, November 2005. Available from: <http://www.london.gov.uk/assembly/reports/environment/derelict-land.pdf>
- London Assembly (2006) A lot to lose: London's disappearing allotments. London Assembly Environmental Report, October 2006. Available from: <http://www.london.gov.uk/assembly/reports/environment.jsp>
- Luniak M (2005) Warsaw. In: Kelcey JG, Rheinwald G (eds) Birds in European cities. Ginster, Germany, pp 389–416
- Macleod R, Barnett P, Clark J, Cresswell W (2006) Mass-dependant predation risk as a mechanism for house sparrow declines? *Biol Lett* 2:43–46. doi:10.1098/rsbl.2005.0421
- Magrath RD (1991) Nestling weight and juvenile survival in the blackbird *Turdus merula*. *J Anim Ecol* 60:335–351
- Marchant JH, Hudson R, Carter SP, Whittington PA (1990) Population trends in British breeding birds. British Trust for Ornithology, Tring
- Marzluff JM, Bowman R, Donnelly R (2001) Avian ecology and conservation in an urbanising world. Kluwer, Boston
- Mason CF (2006) Avian species richness and numbers in the built environment: can new housing developments be good for birds? *Biodivers Conserv* 15:2365–2378. doi:10.1007/s10531-004-1236-4
- McCarthy M (2006) The secret life of sparrows. The Independent, 02 August 2006. Available from: <http://news.independent.co.uk/environment/article1210180.ece>
- McIntyre NE (2000) Ecology of urban arthropods: a review and a call to action. *Ann Entomol Soc Am* 93:825–835. doi:10.1603/0013-8746(2000)093[0825:EOUAAAR]2.0.CO;2
- Mulsow R (2005) Hamburg. In: Kelcey JG, Rheinwald G (eds) Birds in European cities. Ginster, Germany, pp 127–152
- Mulsow R (2006) The birdlife of Hamburg. *Hamburger Avifauna Beitrage Special Edition IOC Hamburg* 2006, pp 45–76
- Muratet A, Machon N, Jiguet F, Moret J, Porcher E (2007) The role of urban structures in the distribution of wasteland flora in the greater Paris area. *Ecosystems* 10:661–671. doi:10.1007/s10021-007-9047-6
- Murgui E (2005) Valencia. In: Kelcey JG, Rheinwald G (eds) Birds in European cities. Ginster, Germany, pp 335–358
- Noble M, Wright G, Dibben C, Smith GAN, McLennan D, Antilla C, Barnes H, Mokhtar C, Noble S, Avenell D, Gardener J, Covizzi I, Lloyd M (2004) Indices of deprivation. Available from: <http://www.communities.gov.uk>
- Office of the Deputy Prime Minister (now Communities and Local Government) (2003). Sustainable communities: building for the

- future. Available from: http://www.communities.gov.uk/pub/872/SustainableCommunitiesBuildingfortheFutureMaindocument_id1139872.pdf
- Paston S (2000) The house sparrow in Norwich during autumn/winter 2000—a population study. *Norfolk Bird Mamm Rep* 34:289–293
- Pauleit S, Ennos R, Golding Y (2005) Modelling the environmental impacts of urban land use and land cover change—a study in Merseyside, UK. *Landsc Urban Plan* 71:295–310. doi:[10.1016/j.landurbplan.2004.03.009](https://doi.org/10.1016/j.landurbplan.2004.03.009)
- Prowse A (2002) The urban decline of the house sparrow. *Br Birds* 95:143–146
- Raven MJ, Noble DG (2006) The breeding bird survey 2005. Available from: <http://www.bto.org/bbs/results/bbsreport.htm>
- Robinson RA, Siriwardena GM, Crick HQP (2005) Size and trends of the house sparrow *Passer domesticus* population in Great Britain. *Ibis* 147:552–562. doi:[10.1111/j.1474-919x.2005.00427.x](https://doi.org/10.1111/j.1474-919x.2005.00427.x)
- Sanderson RF (1996) Autumn bird counts in Kensington Gardens. *Lond Bird Rep* 60:170–176
- Siriwardena GM, Robinson RA, Crick HQP (2002) Status and population trends of the house sparrow, *Passer domesticus*, in Great Britain. In: Crick HQP, Robinson RA, Appleton GF, Clark NA, Rickard AD (eds) Investigation into the causes of decline of starlings and house sparrows in Great Britain. BTO Research Report 290, DEFRA, Bristol. Available from: <http://www.defra.gov.uk/wildlife-countryside/resprog/findings/sparrow/>
- Small EC, Sadler JP, Telfer MG (2002) Carabid beetle assemblages on urban derelict sites in Birmingham, UK. *J Insect Conserv* 6:233. doi:[10.1023/A:1024491111572](https://doi.org/10.1023/A:1024491111572)
- Snow DW, Perrins CM (eds) (1998) *Birds of the Western Palearctic, concise edn, vol 2*. Oxford University Press, UK
- Stastny K, Bejcek V, Kelcey JG (2005) Prague. In: Kelcey JG, Rheinwald G (eds) *Birds in European cities*. Ginsten, Germany, pp 215–242
- Suhonen J (1993) Predation risk influences the use of foraging sites by tits. *Ecology* 74:1197–1203
- Summers-Smith JD (2003) The decline of the house sparrow: a review. *Br Birds* 95:143–146
- Székely T, Szép T, Juhász T (1989) Mixed species flocking by tits (*Parus* spp.): a field experiment. *Oecologia* 78:490–495. doi:[10.1007/BF00378739](https://doi.org/10.1007/BF00378739)
- Sziemer P, Holzer T (2005) Vienna. In: Kelcey JG, Rheinwald G (eds) *Birds in European cities*. Ginsten, Germany, pp 359–388
- Toms M (2006) The BTO/CJ Gardenbirdwatch. Available from: <http://www.bto.org/gbw/>
- Veit RR, Lewis MA (1996) Dispersal, population growth and the Allee effect: dynamics of the house finch invasion of eastern North America. *Am Nat* 148:225–274
- Vincent K (2005) Investigating the causes of the decline of the urban house sparrow *Passer domesticus* in Britain. PhD Thesis, De Montfort University, Leicester. Available from: <http://www.katevincent.org>
- Weiserbs A, Jacob J-P (2005) Brussels. In: Kelcey JG, Rheinwald G (eds) *Birds in European cities*. Ginsten, Germany, pp 81–102
- Whittingham MJ, Evans KL (2004) The effects of habitat structure on predation risk of birds in agricultural landscapes. *Ibis* 146:210–220. doi:[10.1111/j.1474-919X.2004.00370.x](https://doi.org/10.1111/j.1474-919X.2004.00370.x)
- Witt K (2000) Situation of birds in the urban environment: the example of Berlin. *Vogelwelt* 121:107–128
- Witt K (2005) Berlin. In: Kelcey JG, Rheinwald G (eds) *Birds in European cities*. Ginsten, Germany, pp 17–40
- Woods M, McDonald RA, Harris S (2003) Predation by domestic cats *Felis catus* in Great Britain. *Mammal Rev* 33:174–188. doi:[10.1046/j.1365-2907.2003.00017.x](https://doi.org/10.1046/j.1365-2907.2003.00017.x)
- Wotton SR, Field R, Langston RHW, Gibbons DW (2002) Homes for birds: the use of houses for nesting by birds in the UK. *Br Birds* 95:586–592